THE CECH CHAIN COMPLEX OF A FAMILY OF MORPHISMS IS EXACT AWAY FROM DEGREE ZERO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The displayed definition of delta in the proof carries the author's sign (-1)^p where the index i is intended; it is reproduced verbatim and not corrected.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter reference is located at its \label line inside the pinned chapter file; the cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Cech complex of a family and the presheaves Z_X represented by objects. Same chapter, Definition environment labelled definition-cech-complex (source line 1477, Stacks tag 03OL); the statement refers to its notation and assumptions. Supplies the complex Z^bullet_U of abelian presheaves; its exactness away from degree zero is the counted content.
- Simplicial acyclicity of Z[S] <- Z[S x S] <- Z[S x S x S] <- .... Standard homological algebra; the proof exhibits an explicit nullhomotopy for the operator delta rather than citing it. Supplies the left-hand target of the reduction; the reduction itself and the homotopy are retained in the counted span.

Source reference. Original definition 03OL: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 1477–1477. Same-chapter pointer named in the first line of the counted statement (line 1628) as the source of notation and assumptions; the reduction and the explicit homotopy are retained verbatim in the counted span.

Source reference. Original lemma 03AT: sources/stacks-chow/tags.txt, source lines 4322–4322. Cross-chapter 'see for more details' pointer at proof lines 1696-1698, invoked after the homotopy is already constructed; the counted proof is self-contained.

Lemma 0.1. *Notation and assumptions as in Definition 03OL. The complex of abelian presheaves*

$$\mathbf{Z}^\bullet_{\mathcal{U}} \quad : \quad \bigoplus_{i_0 \in I} \mathbf{Z}_{U_{i_0}} \leftarrow \bigoplus_{i_0, i_1 \in I} \mathbf{Z}_{U_{i_0} \times_U U_{i_1}} \leftarrow \bigoplus_{i_0, i_1, i_2 \in I} \mathbf{Z}_{U_{i_0} \times_U U_{i_1} \times_U U_{i_2}} \leftarrow \ldots$$

is exact in all degrees except 0 *in* $PAb(\mathcal{C})$.

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Proof. For any $V \in \mathrm{Ob}(\mathcal{C})$ the complex of abelian groups $\mathbf{Z}^\bullet_{\mathcal{U}}(V)$ is

$$\mathbf{Z}\left[\coprod_{i_0 \in I} \mathrm{Mor}_{\mathcal{C}}(V, U_{i_0})\right] \leftarrow \mathbf{Z}\left[\coprod_{i_0, i_1 \in I} \mathrm{Mor}_{\mathcal{C}}(V, U_{i_0} \times_U U_{i_1})\right] \leftarrow \ldots =$$
$$\bigoplus\nolimits_{\varphi : V \to U} \left(\mathbf{Z}\left[\coprod_{i_0 \in I} \mathrm{Mor}_{\varphi}(V, U_{i_0})\right] \leftarrow \mathbf{Z}\left[\coprod_{i_0, i_1 \in I} \mathrm{Mor}_{\varphi}(V, U_{i_0}) \times \mathrm{Mor}_{\varphi}(V, U_{i_1})\right] \leftarrow \ldots\right)$$

where

$$\mathrm{Mor}_{\varphi}(V, U_i) = \{V \to U_i \text{ such that } V \to U_i \to U \text{ equals } \varphi\}.$$

Set $S_\varphi = \coprod_{i \in I} \mathrm{Mor}_{\varphi}(V, U_i)$, so that

$$\mathbf{Z}^\bullet_{\mathcal{U}}(V) = \bigoplus_{\varphi : V \to U} (\mathbf{Z}[S_\varphi] \leftarrow \mathbf{Z}[S_\varphi \times S_\varphi] \leftarrow \mathbf{Z}[S_\varphi \times S_\varphi \times S_\varphi] \leftarrow \ldots).$$

Thus it suffices to show that for each $S = S_\varphi$, the complex

$$\mathbf{Z}[S] \leftarrow \mathbf{Z}[S \times S] \leftarrow \mathbf{Z}[S \times S \times S] \leftarrow \ldots$$

is exact in negative degrees. To see this, we can give an explicit homotopy. Fix $s \in S$ and define $K : n_{(s_0, \ldots, s_p)} \mapsto n_{(s, s_0, \ldots, s_p)}$. One easily checks that K is a nullhomotopy for the operator

$$\delta : \eta_{(s_0, \ldots, s_p)} \mapsto \sum\nolimits_{i=0}^{p} (-1)^p \eta_{(s_0, \ldots, \hat{s}_i, \ldots, s_p)}.$$

See Cohomology on Sites, Lemma 03AT for more details. □